

# Ali Zamani

Senior Machine Learning Engineer

LLMs | Multi-Agent Systems | RAG | MLOps | Production ML

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## Summary

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Senior ML Engineer, 5+ years in production ML, with the last 3 focused on LLM, RAG, and multi-agent systems on GCP (Vertex AI) across retail, travel, and healthcare. Built a RAG retrieval pipeline serving 10K daily queries at 97% recall, and shipped a GenAI feature that drove a 30% validated via A/B test.

## Relevant Work Experience

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### Senior Machine Learning Engineer (Full-Time)

Jun. 2025 - Present

Loblaw Digital, Toronto, ON

- Architected a **planner-based multi-agent AI platform** with specialized domain agents, multi-layer guardrails, and hierarchical memory (long-term, short-term, episodic), supporting **7 domain agents**.
- Led design and implementation of a production **hybrid-search RAG platform**, making the strategic build-versus-buy call over managed retrieval services. Built a custom retrieval stack on **Vertex AI**, **SPLADE**, **Qdrant**, **Cloud Run**, and **Airflow**, cutting projected infrastructure costs by **\$75K–\$85K annually** while achieving **97% recall** and **78% precision**.
- Designed a reusable **Agent Tool Framework** using **LangGraph** and **MCP** for standardized tool integration, adopted by **2 teams** as the production foundation for **4 AI initiatives**, cutting new-tool integration time from **10 hours to 30 minutes**.
- Architected a **search routing service** using a trained **XGBoost** classifier to route user queries across **3 categories**, in collaboration with **Search** and **Backend** engineering teams, reducing query misroutes by **60%**.

### Machine Learning Engineer (Full-Time) 📺

Apr. 2023 - Jun. 2025

Priceline (Booking Holdings), Toronto, ON

- Built a reusable **ML platform** on **Vertex AI** standardizing preprocessing, training, feature engineering, versioning, backfilling, and online/offline deployment via reusable **Airflow** pipelines, cutting model deployment time from **30 days to 1 day** across **5 use cases**.
- Architected a **GenAI framework** on GCP with CI/CD, production inference, and automated evaluation via **LLM-as-a-Judge**; reusable inference services with rate limiting and parallel execution now support 5 LLM workflows in production.
- Developed and deployed a production **hotel review summarization** solution combining sentiment analysis and LLM-based summarization to surface high-impact customer feedback, resulting in a **30% conversion uplift** validated through A/B testing.

### Data Scientist (Contract Position)

Sept. 2022 - Dec. 2022

AltaML, Calgary, AB

- Built a rock-core permeability prediction model using **LightGBM** and **XGBoost**, achieving **94% accuracy**; adopted by the client to replace manual core analysis, projected to save **\$10M** in costs.

### Data Scientist and Chatbot Developer (MIRA Chatbot - 📺)

Jan. 2021 – Sept. 2022

Department of Computing Science, University of Alberta and Amii, Edmonton, AB

- Built and deployed the full-stack **MIRA chatbot** ([mymira.ca](http://mymira.ca)) end-to-end, from NLP pipeline to production front-end.
- Evaluated multiple **RNN** architectures for intent detection and entity extraction, achieving **97% F1-score** on intent detection and **83% F1-score** on entity extraction.

## Education

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### M.Sc. in Computer Science

University of Alberta – Thesis: MIRA Chatbot, advised by Dr. Osmar R. Zaiane

Jan. 2021 – Aug. 2022

*Edmonton, AB, Canada*

### M.Sc. in Computer Engineering

Amirkabir University of Technology – Thesis: Computer Network Optimization

Sep. 2017 – Sep. 2020

*Tehran, Iran*

### B.Sc. in Electrical Engineering

Kashan University

Sep. 2013 – Sep. 2017

*Isfahan, Iran*

## Technical & Soft Skills

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**Programming:** Python (5+ years), C++, C, JavaScript

**LLM / GenAI / Agents:** LangChain, LangGraph, MCP, prompt engineering, agent orchestration, RAG, LLM-as-a-Judge, multimodal models

**ML / Modeling:** TensorFlow, PyTorch, Hugging Face, Scikit-learn, XGBoost, LightGBM, SPLADE, time series forecasting

**Cloud / Infra:** GCP (Vertex AI, BigQuery, Cloud Storage), Azure, Docker, Kubernetes, Airflow, Cloud Run

**Data / Search:** Qdrant, SQL, PostgreSQL, MySQL, BigQuery, Spark

**Tools:** Git, NGINX, Jira

**Leadership:** Mentored 2 junior engineers, led cross-functional design reviews across 3 teams

## Selected Projects

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### Google Hackathon — 1st Place (2023) and 3rd Place (2022)

- **Neighborhood Recommendation System:** Built a vector-embedding-based recommendation engine matching users to neighborhoods based on travel preferences.
- **Travel Assistance Chatbot:** Built a GPT-4-powered chatbot to help users discover travel destinations conversationally.

### ML-Pipeline Template 🔄

- Developed an ML pipeline template to create a user friendly utility to drastically speed up the development and implementation of a machine learning model for all sorts of various problems.

### Microsoft Hackathon

- **Burnout Detection System:** Built an ML pipeline on Azure using a fine-tuned **BERT** model for sentiment/stress detection in call center conversations, achieving 90% accuracy.

### Spotify Music Recommender 🔄

- Built a music recommendation system using content-based filtering, clustering, and deep learning on the Spotify dataset.

## Selected Publications

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### Developing and Implementing a Mental Health Chatbot to Support Healthcare Workers [[LINK](#)]

2021, A.Zamani, M.Gharayat, J.Nobel, O.Zaiane, E.Stroulia – REMAP-D, Vancouver, British Columbia, Canada

### An Efficient Load Balancing Approach for Service Function Chain Mapping [[LINK](#)]

2021, Ali Zamani, Bahador Bakhshi, Saeed Sharifian